

Topology and Manifolds

MH3600

Chapter 5: Connectedness

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Outline

- 1 Connected Spaces
- 2 Connectedness in the real line. Path connectedness.
- 3 Components and local connectedness
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- 1 Connected Spaces
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Connectedness

Definition

Let X be a topological space. A **separation of X** is a pair U, V of disjoint nonempty open subsets of X whose union is X .

Space X is **connected** if there is no separation of X .

Alternative definition

An alternative definition is that X is **connected** if and only if the only subsets of X that are both open and closed in X are the empty set and X itself.

Indeed, If A is a nonempty proper subset of X that is both open and closed in X , then the sets $U = A$ and $V = X - A$ constitute a separation of X , for they are open, disjoint, and nonempty, and their union is X . Conversely, if U and V form a separation of X , then U is nonempty and different from X , and it is both open and closed in X .

Connectivity is a topological property

Since connectivity is defined in term of open sets only, if X is connected and Y is homeomorphic to X then Y is connected. That is, connectivity is a topological property.

Connectedness for subspaces

Lemma

Let Y be a subspace of X . Two disjoint nonempty sets A and B whose union is Y form a separation of Y if and only if A contains no limit points of B and B contains no limit points of A .

Example

Let $Y = [-1, 0) \cup (0, 1] \subset \mathbb{R}$. Then $A = [-1, 0)$ and $B = (0, 1]$ form a separation of Y and so Y is not connected.

Example

The rationals $\mathbb{Q}$ are not connected. For any $a \in \mathbb{R} \setminus \mathbb{Q}$, a separation of $\mathbb{Q}$ is given by $A = \mathbb{Q} \cap (-\infty, a)$ and $B = \mathbb{Q} \cap (a, \infty)$.

New connected spaces out of old ones

We now give several results on how to build new connected sets from known connected sets. Since showing a set is connected is equivalent to showing that there is not a separation, the proofs are by contradiction.

Lemma

If sets C and D form a separation of X and if Y is a connected subspace of X , then Y lies entirely in either C or in D .

Proof

Since C and D are both open in X , the sets $C \cap Y$ and $D \cap Y$ are open in Y . These two sets are disjoint and their union is Y . Assume, by contradiction, that both are nonempty. Then these two sets form a separation of Y . Contradicting the hypothesis that Y is connected. So either $C \cap Y$ or $D \cap Y$ is an empty set and so Y lies entirely in either C or in D . ■

Theorem

The *union of a collection of connected subspaces of X that have a point in common* is *connected*.

Proof

Let $\{A_\alpha\}$ be a collection of subspaces of a space X . Let p be a point in $\cap A_\alpha$. Let $Y = \cup A_\alpha$ and assume $Y = C \cup D$ where C and D are a separation of Y . Point p must be in either C or D . Without loss of generality, say $p \in C$. Since A_α is connected, it must lie entirely in either C or in D , by the preceding Lemma. It cannot lie in D since $p \in A_\alpha$ and $p \in C$. Hence $A_\alpha \subset C$ for every α . So $Y = \cup A_\alpha \subset C$. But this contradicts the fact that D is nonempty. So the assumption that there is a separation of Y is false and hence the union $Y = \cup A_\alpha$ is connected.



Theorem

Let A be a *connected subspace of X* . If $A \subset B \subset \bar{A}$, then B is also *connected*.

Proof

Let A be connected and let $A \subset B \subset \bar{A}$. Assume that $B = C \cup D$ where C and D are a separation of B . By the preceding Lemma, the set A must lie entirely in C or in D . Without loss of generality, suppose $A \subset C$. Then $\bar{A} \subset \bar{C}$. Since $\bar{C}$ and D are disjoint by Lemma on slide 6, then $B \cap D \subset \bar{A} \cap D \subset \bar{C} \cap D = \emptyset$. But this contradicts the fact that as part of a separation, D is a nonempty subset of B . So the assumption that a separation of B exists is false, and so B is connected. ■

Connectedness and continuity

The whole point of calling a function “continuous” is not because of ε 's and δ 's or because of inverse images of open sets, but because **continuous functions take connected sets to connected sets**. This property is proved in the next theorem.

Theorem

The image of a connected space under a continuous map is connected.

Proof of the Theorem

Let $f : X \rightarrow Y$ be a continuous function where X is connected. Let $Z = f(X)$. Since the map obtained from f by restricting its range to the space Z is also continuous (by Chap 2), without loss of generality we can assume that f is surjective (onto).

Assume $Z = A \cup B$, where A and B form a separation of Z . Then $f^{-1}(A)$ and $f^{-1}(B)$ are disjoint sets (since A and B are disjoint) whose union is X , which are open since f is continuous, and which are nonempty since f is onto. Therefore $f^{-1}(A)$ and $f^{-1}(B)$ are a separation of X . But this contradicts the hypothesis that X is connected.

Hence there is no separation of $Z = f(X)$ and so $f(X)$ is connected.



Connectedness and Products

Theorem

A *finite Cartesian product of connected spaces* is *connected*.

Proof

We prove the result for two connected spaces X and Y and then the general result follows by induction.

Choose $(a, b) \in X \times Y$. Then $X \times \{b\}$ and $\{x\} \times Y$ are connected (for each $x \in X$), as will be shown in the homework. For each $x \in X$, define $T_x = (X \times \{b\}) \cup (\{x\} \times Y)$. Then T_x is connected by Theorem on slide 8 (since (x, b) is in each constituent space). Next, consider

$\bigcup_{x \in X} T_x = X \times Y$. This union is connected by Theorem on slide 8 since the point (a, b) is common to each T_x . That is, $X \times Y$ is connected.

The proof for any finite product of connected spaces follows by induction along with the fact (See Tutorial 5) that $X_1 \times X_2 \times \cdots \times X_n$ is homeomorphic with $(X_1 \times X_2 \times \cdots \times X_{n-1}) \times X_n$. ■

Even though the last theorem holds for finite Cartesian products, the following example shows that **it does not hold** for a (countable) infinite product of connected spaces.

Example ($\mathbb{R}^\omega$ with the box topology is not connected)

Consider the Cartesian product $\mathbb{R}^\omega$ with the **box topology**. Then $\mathbb{R}^\omega$ is the union of set A consisting of all bounded sequences of real numbers and the set B of all unbounded sequences. These sets are disjoint. Let $\mathbf{a} = (a_1, a_2, \dots) \in \mathbb{R}^\omega$. Consider basis element (and hence open set) for the box topology $U_{\mathbf{a}} = (a_1 - 1, a_1 + 1) \times (a_2 - 1, a_2 + 1) \times \dots$.

– If $\mathbf{a} \in A$ is bounded then $U_{\mathbf{a}}$ is an open set containing $\mathbf{a}$ and consisting only of bounded sequences, hence $U_{\mathbf{a}} \subset A$. So A is open.

– If $\mathbf{a} \in B$ is unbounded then $U_{\mathbf{a}}$ is an open set containing $\mathbf{a}$ and consisting only of unbounded sequences, hence $U_{\mathbf{a}} \subset B$. So B is open.

Therefore A and B form a separation of $\mathbb{R}^\omega$ and so $\mathbb{R}^\omega = \mathbb{R} \times \mathbb{R} \times \dots$ **under the box topology is not connected** even though (as will be proved later on) $\mathbb{R}$ is connected.

Example ($\mathbb{R}^\omega$ with the product topology is connected)

Consider $\mathbb{R}^\omega$ with the product topology. Let $\tilde{\mathbb{R}}^n$ denote the subspace of $\mathbb{R}^\omega$ consisting of all sequences $\mathbf{x} = (x_1, x_2, \dots)$ such that $x_i = 0$ for $i > n$. The space $\tilde{\mathbb{R}}^n$ is clearly homeomorphic to $\mathbb{R}^n$ and so (assuming $\mathbb{R}$ is connected) is connected by Theorem on slide 12 ($\tilde{\mathbb{R}}^n \simeq \mathbb{R}^n$ does have the subspace topology induced by $\mathbb{R}^\omega$ with product topology (also true with box!) hence this Theorem applies). The space

$$\mathbb{R}^\infty = \{\mathbf{x} = (x_1, x_2, \dots) \in \mathbb{R}^\omega \mid x_i \neq 0 \text{ for only finitely many } i \in \mathbb{N}\}$$

can be written as the union $\mathbb{R}^\infty = \bigcup_{n \in \mathbb{N}} \tilde{\mathbb{R}}^n$, and since $\mathbf{0} \in \tilde{\mathbb{R}}^n$ for all $n \in \mathbb{N}$, then $\mathbb{R}^\infty$ (under the product (and box) topology) is connected by Theorem on slide 8.

We have $\overline{\mathbb{R}^\infty} = \mathbb{R}^\omega$ with respect to the product topology (Tutorial 5, $\overline{\mathbb{R}^\infty} = \mathbb{R}^\infty$ with box). Hence we are in the situation $\mathbb{R}^\infty \subset \mathbb{R}^\omega \subset \overline{\mathbb{R}^\infty}$ with $\mathbb{R}^\infty$ connected. From Theorem on slide 9, $\mathbb{R}^\omega$ is connected.

Ok, but why $\mathbb{R}^\omega = A \cup B$ is no more a separation?

What about arbitrary products?

The argument given in this example extends from a countable product of copies of $\mathbb{R}$ (under the product topology) to an arbitrary product of connected spaces under the product topology (see Exercises).

This implies that **arbitrary product of connected spaces** are **connected** when we use the **product topology**.

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Connectedness in continua

In this section we shall state that $\mathbb{R}$ is connected, and so are the intervals and rays in $\mathbb{R}$.

Applications: generalized intermediate value theorem and balls and spheres in euclidean space are connected.

Proof involves a new notion, called path connectedness.

The fact that intervals and rays in $\mathbb{R}$ are connected does not depend on the algebraic properties of $\mathbb{R}$, but only on its order properties. To make this clear, we shall state the theorem for an arbitrary ordered set that has the order properties of $\mathbb{R}$. Such a set is called a linear continuum.

Definition

An ordered set L having more than one element is called a linear continuum if the following hold:

- (1) L has the least upper bound property.
- (2) If $x < y$, there exists z such that $x < z < y$.

Order relation: Comparability, Nonreflexivity, Transitivity

L has the least upper bound property: every subset $L_0 \neq \emptyset$ of L bounded above has a least upper bound (counter-ex: $\mathbb{Q}$).

Theorem

If L is a linear continuum in the order topology, then L is connected, and so are intervals and rays in L .

Applying this Theorem to the linear continuum $\mathbb{R}$, we get the following result.

Corollary

The real line $\mathbb{R}$ is connected and so are intervals and rays in $\mathbb{R}$.

The following is a familiar result from Calculus but in the setting of connected topological spaces and ordered topological spaces.

Theorem (Intermediate Value Theorem)

Let $f : X \rightarrow Y$ be a *continuous map*, where X is a *connected space* and Y is an *ordered set in the order topology*. If a and b are two points of X and if r is a point of Y lying between $f(a)$ and $f(b)$, then *there exists a point $c \in X$ such that $f(c) = r$.*

Proof of the Intermediate Value Theorem

Suppose f , X , and Y are as hypothesized, and r lying between $f(a)$ and $f(b)$.

The sets $A = f(X) \cap (-\infty, r)$ and $B = f(X) \cap (r, +\infty)$ are disjoint (since $(-\infty, r)$ and $(r, +\infty)$ are disjoint) and nonempty since $f(a)$ is in one of these sets and $f(b)$ is in the other. Each is open in $f(X)$ under the subspace topology.

Note that we have the disjoint union $f(X) = A \cup (f(X) \cap \{r\}) \cup B$. Assume there is no point $c \in X$ such that $f(c) = r$. Then $f(X) = A \cup B$ and A and B form a separation of $f(X)$. But since X is connected and f is continuous then $f(X)$ is connected by Theorem on slide 10, a contradiction. So the assumption that there is no such $c \in X$ is false and hence $f(c) = r$ for some $c \in X$. ■

Example (A linear continuum different from $\mathbb{R}$)

The **ordered square**, I_o^2 , is $I \times I = [0, 1] \times [0, 1]$ with the dictionary order topology. This forms a linear continuum as well. It satisfies the least upper bound property for the following reasons:

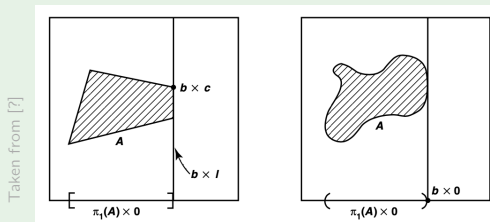
Let $A \subset I \times I$. Let $\pi_1 : I \times I \rightarrow I$ be the projection into the first coordinate. Let $b = \sup(\pi_1(A))$, ($\pi_1(A) \subset I$ is a subset of $\mathbb{R}$ and so satisfies the least upper bound property; notice that 1 is an upper bound of $\pi_1(A)$). Now there are two options (see the figure next slide):

- If $b \in \pi_1(A)$ then A intersects $\{b\} \times I$ (see the figure). Since $\{b\} \times I$ is homeomorphic to I , then $A \cap (\{b\} \times I) \subset \{b\} \times I$ has upper bound $(b, 1)$, and so has the least upper bound of the form (b, c) , for some $c \leq 1$. (b, c) is also the least upper bound for A .

Example (Continued)

- If $b \notin \pi_1(A)$ then $(b, 0)$ is the least upper bound of A because no element of the form (b', c) with $b' < b$ can be an upper bound for A , for then b' would be an upper bound for $\pi_1(A)$, contradicting the fact that $b > b'$ is the least upper bound of $\pi_1(A)$. So the ordered square has the least upper bound property.

The property that $x < y$ implies the existence of z with $x < z < y$ is easy and left as an exercise.



Path connectedness

Definition

Given points x and y of the space X , a **path** in X from x to y is a continuous map $f : [a, b] \rightarrow X$ of some closed interval in the real line into X , such that $f(a) = x$ and $f(b) = y$. A space X is said to be **path connected** if every pair of points of X can be joined by a path in X .

A path connected space X is connected

Indeed, suppose X is not connected and let $X = A \cup B$ be a separation of X . Let $f : [a, b] \rightarrow X$ be any path in X . Being the continuous image of a connected set, the set $f([a, b])$ is connected, so that it lies entirely in either A or B (by Lemma slide 7). Therefore, there is no path in X joining a point of A to a point of B , contrary to the assumption that X is path connected.

The converse does not hold: a **connected space need not be path connected**.

Example (Balls)

Define the **unit ball** B^n in $\mathbb{R}^n$ by the equation

$$B^n = \{\mathbf{x} \mid \|\mathbf{x}\| \leq 1\}$$

where

$$\|\mathbf{x}\| = \|(x_1, \dots, x_n)\| = (x_1^2 + \dots + x_n^2)^{1/2}$$

The **unit ball** is **path connected**: given any two points $\mathbf{x}$ and $\mathbf{y}$ of B^n , the straight-line path $f : [0, 1] \rightarrow \mathbb{R}^n$ defined by $f(t) = (1 - t)\mathbf{x} + t\mathbf{y}$ lies in B^n . For if $\mathbf{x}$ and $\mathbf{y}$ are in B^n and t is in $[0, 1]$,

$$\|f(t)\| \leq (1 - t)\|\mathbf{x}\| + t\|\mathbf{y}\| \leq 1.$$

When **path connectedness** can be **achieved with a straight line** we say that the space is **convex**.

A similar argument shows that **every open ball** $B_d(\mathbf{x}, \epsilon)$ and every closed ball $\overline{B_d(\mathbf{x}, \epsilon)}$ in $\mathbb{R}^n$ is **path connected** and **convex**.

Example (The punctured Euclidean space)

Define the **punctured Euclidean space** to be the space $\mathbb{R}^n - \{\mathbf{0}\}$, where $\mathbf{0}$ is the origin in $\mathbb{R}^n$. If $n > 1$, this space is **path connected**: Given $\mathbf{x}$ and $\mathbf{y}$ different from $\mathbf{0}$, we can join $\mathbf{x}$ and $\mathbf{y}$ by the straight-line path between them if that path does not go through the origin. Otherwise, we can choose a point $\mathbf{z}$ not on the line joining $\mathbf{x}$ and $\mathbf{y}$, and take the broken-line path from $\mathbf{x}$ to $\mathbf{z}$, and then from $\mathbf{z}$ to $\mathbf{y}$. This shows that $\mathbb{R}^n - \{\mathbf{0}\}$ is **path connected** but **not convex**.

Example (Unit spheres)

Define the **unit sphere** S^{n-1} in $\mathbb{R}^n$ by the equation

$$S^{n-1} = \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x}\| = 1\}.$$

If $n > 1$, it is **path connected**. Indeed, the map $g : \mathbb{R}^n - \{0\} \rightarrow S^{n-1}$ defined by $g(\mathbf{x}) = \mathbf{x}/\|\mathbf{x}\|$ is continuous and surjective; and it is easy to show that **the continuous image of a path connected space is path connected**.

Example (The topologist's sine curve)

Let S denote the following subset of the plane.

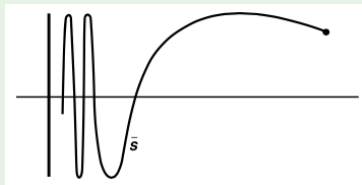
$$S = \{(x, \sin(1/x)) \mid 0 < x \leq 1\} \subset \mathbb{R}^2.$$

Because S is the image of the connected set $(0, 1]$ under a continuous map, S is connected. Therefore, its closure $\bar{S}$ in $\mathbb{R}^2$ is also connected. We have

$$\bar{S} = \{0\} \times [-1, 1] \cup S.$$

The set $\bar{S}$ is called the **topologist's sine curve**. We will show that $\bar{S}$ is **not path connected**.

Taken from [?]



Example (The topologist's sine curve (I))

Suppose there is a path $f : [a, c] \rightarrow \bar{S}$ beginning at the origin and ending at a point of S . The set of those t for which $f(t) \in \{0\} \times [-1, 1]$ is closed, so it has a largest element b . Then $f : [b, c] \rightarrow \bar{S}$ is a path that maps b somewhere into the vertical interval $\{0\} \times [-1, 1]$ and maps the other points of $[b, c]$ to points of S .

Let us write $f(t) = (x(t), y(t))$. Then $x(b) = 0$ (since $f(b) \in \{0\} \times [-1, 1]$) and $x(t) > 0$ for $t > b$ (since $f((b, c]) \subset S$), and $y(t) = \sin(1/x(t))$ for $t > b$.

Construct a sequence $\{t_n\}$ as follows: For $n \in \mathbb{N}$, we can choose u_n such that $x(b) = 0 < u_n < x(b + 1/n)$ and $\sin(1/u_n) = (-1)^n$.

Since $x(t)$ is continuous, by the **Intermediate Value Theorem** there is some t_n with $b < t_n < b + 1/n$ such that $x(t_n) = u_n$. But then $\{t_n\} \rightarrow b$ and $f(t_n) = (x(t_n), \sin(1/x(t_n))) = (x(t_n), \sin(t_n/u_n)) = (x(t_n), (-1)^n)$. so $\{f(t_n)\}$ does not converge which contradicts the continuity of f .

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Connected components

Definition

Given topological space X , define an equivalence relation on X by setting $x \sim y$ if there is a connected subspace of X containing both x and y . The equivalence classes are called **connected components** of X .

$\sim$ is an equivalence relation

Symmetry and **reflexivity** are clear. **Transitivity** follows from the fact that if A is connected and contains x and y , and B is connected and contains y and z , then $A \cup B$ contains x and z and $A \cup B$ is connected by Theorem on slide 8 (since both A and B contain y). Therefore, $x \sim y$ and $y \sim z$ implies $x \sim z$.

Theorem

The *connected components* of X are *connected* and *disjoint* subspaces of X *whose union is X* . Moreover, each nonempty connected subspace of X intersects only one of them.

Proof

Let us call them just “components” so far.

Since the components are by definition equivalence classes, then the components are disjoint and their union gives X (we have seen that equivalence classes on a set form a partition of the set).

Assume that a connected subspace A of X intersects two disjoint nonempty components C_1 and C_2 , say at x_1 and x_2 , respectively. Then $x_1 \sim x_2$ since they both belong to A connected, so $C_1 = C_2$, a contradiction since we assumed them disjoint. So the assumption that a connected subspace can intersect two components is false.

Proof (continued)

To show that a connected component C is connected, let $x_0 \in C$. Then for each $x \in C$ we have $x_0 \sim x$, so there is a connected subspace A_x containing x_0 and x . From the previous result, a connected subspace cannot intersect two components and so $A_x \subset C$. Therefore, $C = \bigcup_{x \in C} A_x$. Since each A_x is connected and $x_0 \in A_x$ for all $x \in C$ then by Theorem on slide 8, C is connected. ■

Path connected components

Definition

Define another equivalence relation on X as $x \sim y$ if and only if there is a path in X from x to y . The equivalence classes are called **path connected components of X** .

$\sim$ is an equivalence relation

$\sim$ is **reflexive** since a constant path $f(t) = x$ is a path from x to x . For **symmetry**, if $x \sim y$ because $f : [0, 1] \rightarrow X$ is a path from x to y , then $g(t) = f(1 - t)$ is a path from y to x and so $y \sim x$.

Before we work on the **transitivity** we note that if there is a path $f : [a, b] \rightarrow X$ from x to y then $g : [c, d] \rightarrow X$ defined as $g(t) = f((t(b - a) + ad - bc)/(d - c))$ is a path from x to y with domain $[c, d]$. Now, if $x \sim y$ and $y \sim z$ then there is a path $f : [0, 1] \rightarrow X$ from x to y and a path $g : [1, 2] \rightarrow X$ from y to z (using the comment above we can assume the given domains of f and g). So by the **Pasting Lemma**

$$h(t) = \begin{cases} f(t) & \text{for } t \in [0, 1] \\ g(t) & \text{for } t \in (1, 2] \end{cases}$$

is a path from x to z with domain $[0, 2]$; therefore $x \sim z$ and $\sim$ is transitive. So $\sim$ **is an equivalence relation**.

Theorem

The *path connected component* of X are *path connected* and *disjoint subspaces* of X *whose union is X* . Moreover, each nonempty path connected subspace of X intersects only one of the path components.

Are components open/close?

Proposition

*Each **connected component** of a space X is **closed**. If X has only **finitely many connected components**, then **each component of X is also open**.*

In general, the **connected components** of a space **may not be open** (see examples below).

Proof

Let C be a connected component of X . By Theorem on slide 9, $\bar{C}$ is also connected. Since the connected components are disjoint by Theorem on slide 30, then $C = \bar{C}$ and so C is closed. If X has only finitely many components then the complement of a component C is a finite union of closed sets by the first part of this lemma, and hence is closed. Hence C is open. ■

Example

We consider $\mathbb{Q} \subset \mathbb{R}$ with the subspace topology.

It is easy to see that each **connected component of $\mathbb{Q}$ is a singleton**. It is closed (as it should) since $\mathbb{Q} \setminus \{q\} = (\mathbb{Q} \cap (-\infty, q)) \cup (\mathbb{Q} \cap (q, \infty))$ is open.

So **each connected component of $\mathbb{Q}$ is closed and not open**.

Example

The **topologist's sine curve**, $\bar{S}$ **is connected**, as shown above. So $\bar{S}$ has one connected component. It was also shown that $\bar{S}$ **is not path connected**. However, $S = \{(x, \sin(1/x)) \mid x \in (0, 1]\}$ is the image of $(0, 1]$ under a continuous function and so is path connected. Also, $V = \{0\} \times [-1, 1]$ is similarly path connected. So $\bar{S}$ **has two path connected components, S and V** . Notice that S **is open** (in the subspace topology of $\bar{S}$) but **not closed** (it does not include its limit points in V - no contradiction since it is not a connected component). V **is closed but not open**.

Local connectedness

Definition

A space X is **locally connected at x** if for every nbd U of x , there is a connected nbd V of x contained in U . If X is locally connected at each of its points, it is said to be **locally connected**. Similarly, a space X is **locally path connected at x** if for every nbd U of x , there is a path connected nbd V of x contained in U . If X is locally path connected at each of its points, it is said to be **locally path connected**.

Example

Each interval in $\mathbb{R}$ is both connected and locally connected. The subspace $[-1, 0) \cup (0, 1] \subset \mathbb{R}$ is not connected but it is locally connected. The topologist's sine curve, $\bar{S} \subset \mathbb{R}^2$, is connected but not locally connected because a nbd (in the subspace topology) U of a point $x \in V = \{0\} \times [-1, 1] \setminus \{(0, 0)\}$ only contains nbd V of x which are not connected because they contain “pieces” of set S (hence separations). The rationals $\mathbb{Q}$ are neither connected nor locally connected.

The following results classify locally connected and path connected spaces.

Theorem

A space X is *locally connected* if and only if for *every open set U of X , each connected component of U is open in X .*

Proof

Suppose X is locally connected. Let U be an open set in X , let C be a connected component of U , and let $x \in C$. Then by the definition of locally connected, there is a connected neighborhood V of x with $V \subset U$. Since V is connected, by Theorem on slide 30, it must lie entirely in the connected component C , i.e., $V \subset C$. So C is open. Conversely, suppose that the connected components of open sets in X are open. Let $x \in X$ and let U be an arbitrary neighborhood of x . Let C be the connected component of U which contains x . Now C is connected and, by hypothesis, open in X . So, by definition, X is locally connected. ■

The proof of the following theorem is similar to the previous one

Theorem

A space X is *locally path connected* if and only if for *every open set U of X , each path component of U is open in X .*

The following result gives the relationship between path connected components and connected components.

Theorem

If X is a topological space, *each path connected component of X lies in a connected component of X . If X is locally path connected, then the connected components and the path connected components are the same.*

Proof of the theorem

Let C be a connected component of X . Let $x \in C$. Let P be the path connected component of X containing x . By the result stated in page 23, P is connected and so $P \subset C$ by Theorem on slide 30, and the first claim holds.

Suppose X is locally path connected. Assume $P \subsetneq C$. Let Q denote the union of all the path components of X that are different from P and which intersect C (since $P \neq C$ then $Q \neq \emptyset$). As above, by the result stated in slide 23 (path connected is connected) and Theorem on slide 30 (connected subspaces intersects only one connected component), each of these path connected components must be in the connected component C . Since the path components partition X by Theorem on slide 34, then the path components in Q , along with the path component P , form a partition of C : $C = P \cup Q$.

Since X is hypothesized to be locally path connected, then by Theorem on slide 39 each path component of X is open in X .

Proof of the theorem (continued)

Therefore P (a path component of X) and Q (a union of path components) are both open in X and are disjoint by construction, so P and Q form a separation of C , a contradiction. So the assumption that $P \neq C$ is false and we have $P = C$. So every connected component of X is a path component and, since the connected components partition X by Theorem on slide 30, conversely every path component is a connected component. ■

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